PROENGINEERING | INDUSTRIAL AI & MEGACAPEX

The Category-of-One Engineering Brand Playbook & Commissioning Kit

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1. BRAND FOUNDATION: THE CATEGORY-OF-ONE POSITIONING

OFFICIAL INSTAGRAM BIO FORMULA (PASTE DIRECTLY):

· 23+ Yrs Commissioning 33kV Substations & Megaprojects (L&T, ABB, PMP)
· Agentic AI for Primavera P6 Controls, Relay Testing & Reliability
· Free Pre-Bid Checklists, SLD Audits & Tools: proengineering.in

2. 5-DAY VIRAL LAUNCH SEQUENCE & HIGH-RETENTION FORMATS

Day & Pillar	Format & Production Asset	Keyword Trigger	Strategic Value
Day 1: Forensic Teardown	Reel #1: 4-Step Animated AI Spec Trap	Comment SPEC for Checklist	Targets EPC Directors losing margin on Division 26 spec traps.
Day 2: Field Commissioning	Reel #2: 33kV Substation Zero-Trip Protocol	Comment SLD for Matrix	High-save checklist: OLTC, CT knee-points, OMICRON injection.
Day 3: Project Controls AI	Reel #3: Primavera P6 Float Trap Audit	Comment P6 for Playbook	Demonstrates AI detecting hidden lags on critical path milestones.
Day 4: Pattern-Interrupt	Reel #4: 2-Character Site Manager vs Architect	Comment RELAY for Protocol	Pattern interrupt: Day-1 soak trip & 2nd harmonic inrush fix.
Day 5: Visual Cheat Sheet	Carousel: Generator Protection on 1 Page	Comment GEN for High-Res	High share potential: ANSI 87G, 40, 46, 64R relay matrix.

3. THE AUTOMATED COMMENT-TO-CLIENT FUNNEL

Step 1: Algorithmic Velocity

Every Reel asks for a specific 1-word comment (SPEC, SLD, P6, RELAY). This spikes comment velocity within the first 60 minutes, signaling high engagement to the Instagram algorithm.

Step 2: Instant Technical Asset DM

Commenters instantly receive a direct link to download this high-density engineering PDF checklist, establishing unquestioned technical authority in under 30 seconds.

Step 3: Executive Advisory Bridge

The final page of every tool invites EPC Project Directors and Plant VPs to book a 30-minute Pre-Bid Spec Audit or Substation RCA Advisory session at proengineering.in.

THE 33 kV SUBSTATION & EPC PRE-BID DIAGNOSTIC CHECKLIST

Field-Tested Across 23+ Years of Megaproject Execution & Commissioning Sign-Offs

Confidential Field Checklist | ProEngineering Technical Practice Document

PART 1: PRE-BID SPECIFICATION & DRAWING AUDIT (STOP THE $400K CHANGE ORDERS)

[ ] Order of Precedence Clause Audit

Identify contradictions between Division 26 spec clauses, BOQ line items, and Tender SLD. Ensure spec supersedes BOQ or submi

[ ] 33kV & 11kV Fault Level Withstand Duration

Verify 31.5 kA vs 40 kA fault ratings for 3 seconds across busbars, CTs, and VCBs. Standard commercial switchgear is rated 1

[ ] CT Class PS Knee-Point (Vk) Calculation

Audit differential 87T CT specifications: Vk >= 2 * If * (Rct + 2*RL + Rr). Reject ambiguous 'standard class' designations be

[ ] SCADA Gateway & IEC 61850 Protocol Interoperability

Verify whether client requires PRP/HSR redundant ring protocols or proprietary vendor relays (MiCOM / ABB Relion / Siemens SI

[ ] Two-Way Cable Takeoff & Trench Derating

Cross-check route length from SLD vs Site Plot Plan. Factor in 15% routing slack plus depth/thermal resistivity derating fact

PART 2: ZERO-TRIP 33 kV ENERGIZATION & COMMISSIONING PROTOCOL

[ ] Dyn11 Vector Group & CT Secondary Loop Polarity

Confirm primary-to-secondary phase displacement (30° lag) and secondary star-point orientation to eliminate false 87T differe

[ ] 15% 2nd-Harmonic Inrush Restraint Setting

Activate 2nd-harmonic restraint threshold (12-15%) on MiCOM P643/ABB RET670 to prevent nuisance tripping during initial trans

[ ] Secondary Injection with OMICRON CMC 356

Perform 3-phase ramp test to verify pick-up, dual-slope characteristic (Slope 1 @ 20-30%, Slope 2 @ 70-80%), and high-set ins

[ ] Vacuum Circuit Breaker (VCB) Timing & Contact Resistance

Measure breaker closing time (< 60ms), opening time (< 45ms), and pole-to-pole contact resistance (< 35 micro-ohms) using 100

[ ] 24-Hour No-Load Soak & Online DGA Oil Sample

Maintain 33kV energization for 24 hours under continuous vibration and acoustic inspection. Pull post-soak transformer oil sa

NEED AN EXPERT PRE-BID SPEC AUDIT OR COMMISSIONING RCA?

ProEngineering partners with EPC Contractors, Project Directors, and Industrial Plant Heads
to eliminate multi-crore change orders and optimize Primavera P6 project execution via Industrial AI.

Book a Confidential Advisory Diagnostic: https://proengineering.in | contact@proengineering.in